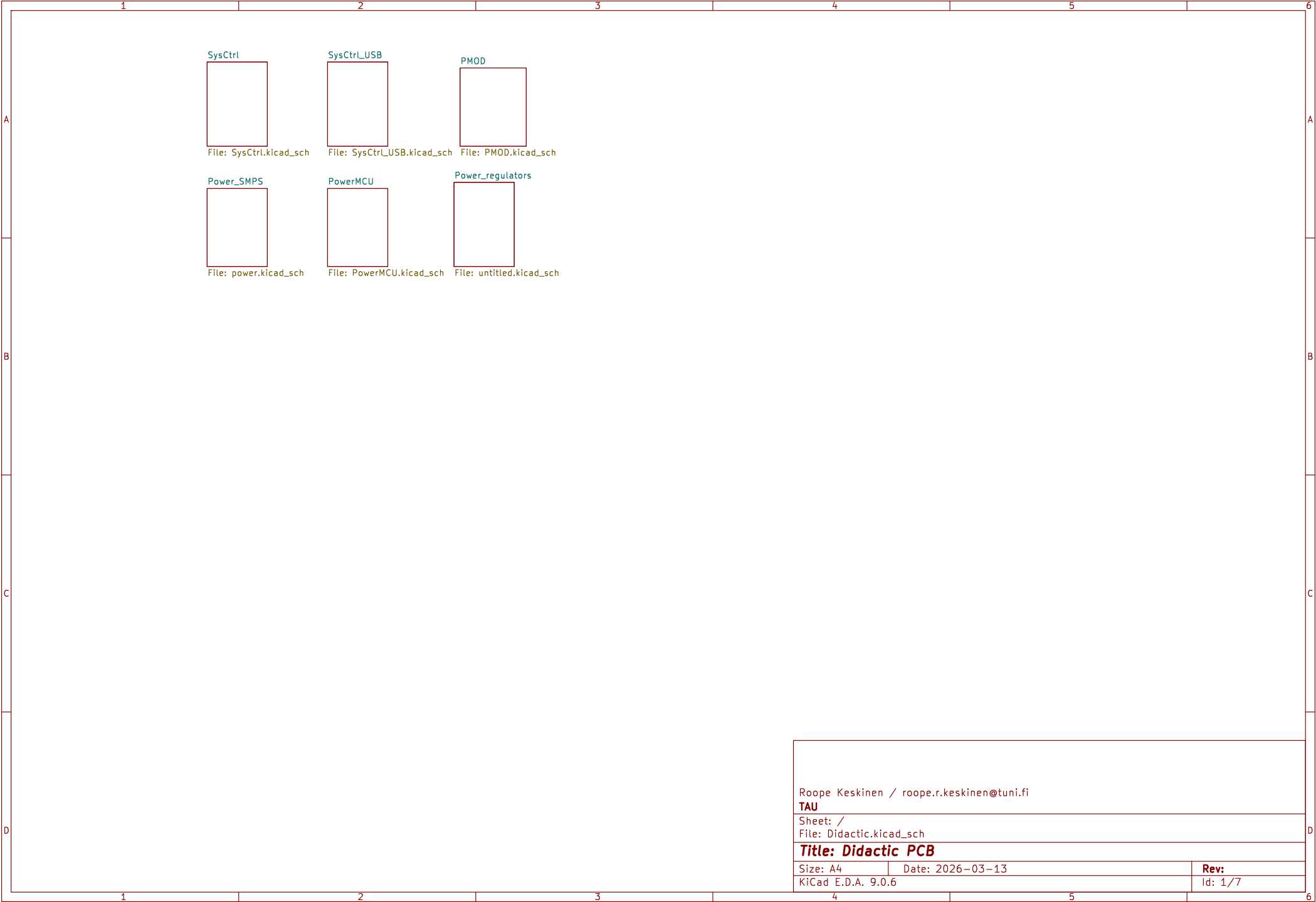
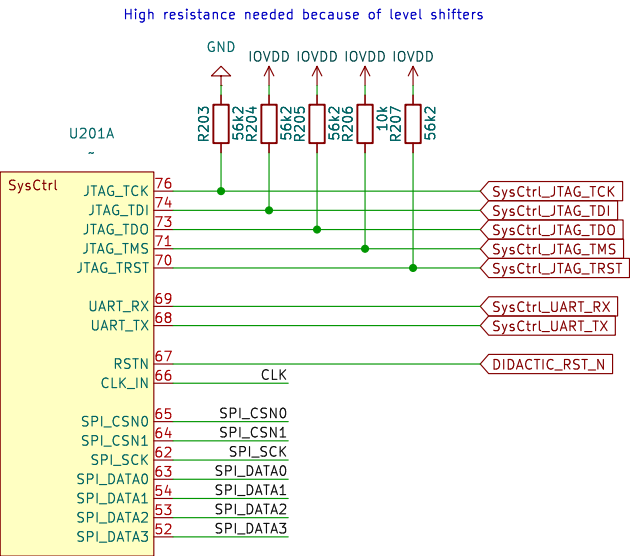
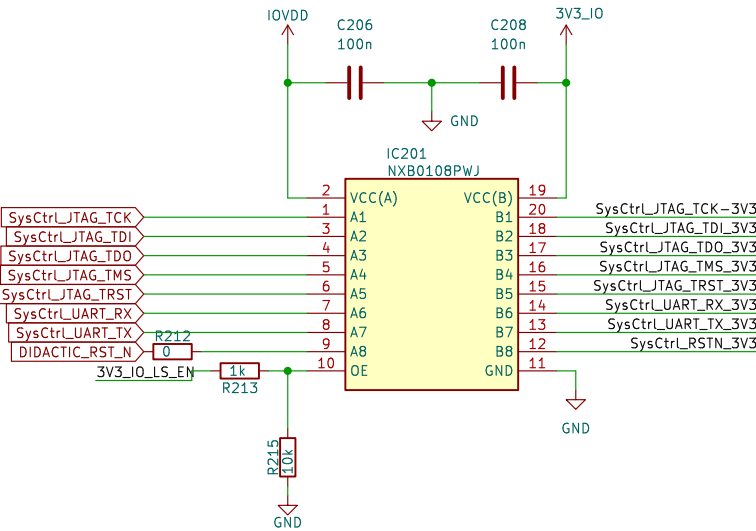




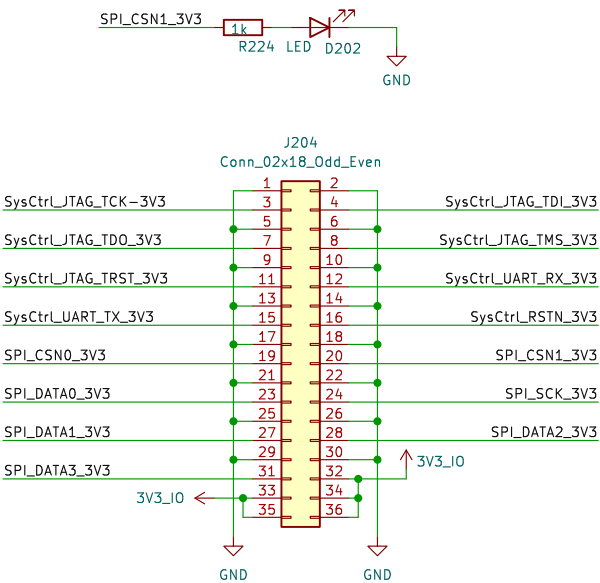
SysCtrl



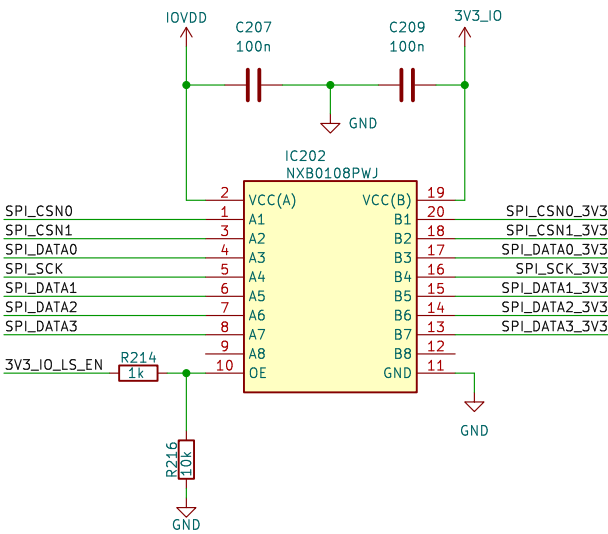
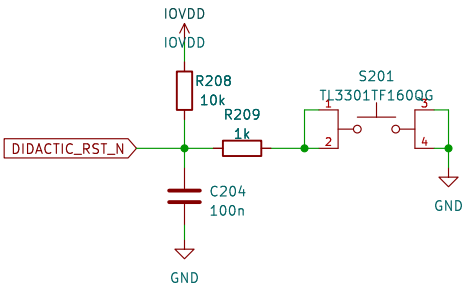
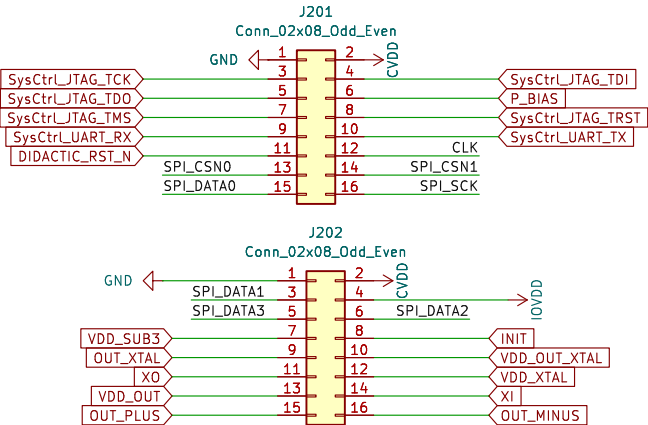
1.8V/3.3V Level shifters



SysCtrl 3.3V IO

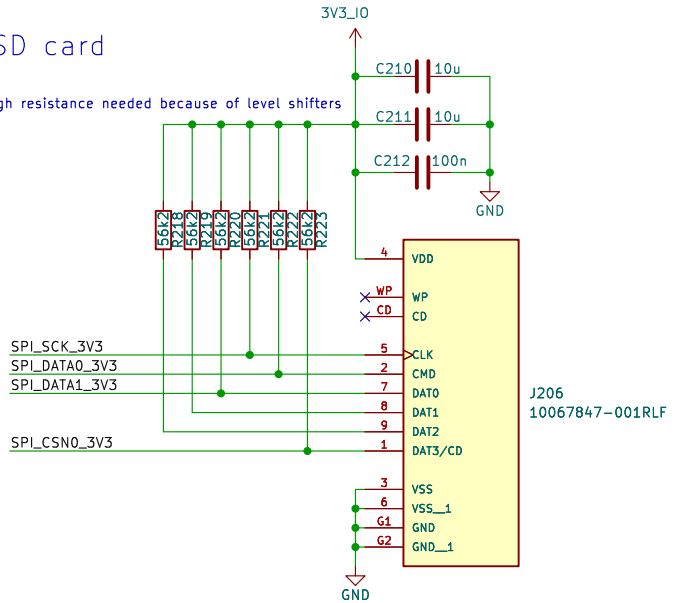


1.8V IO

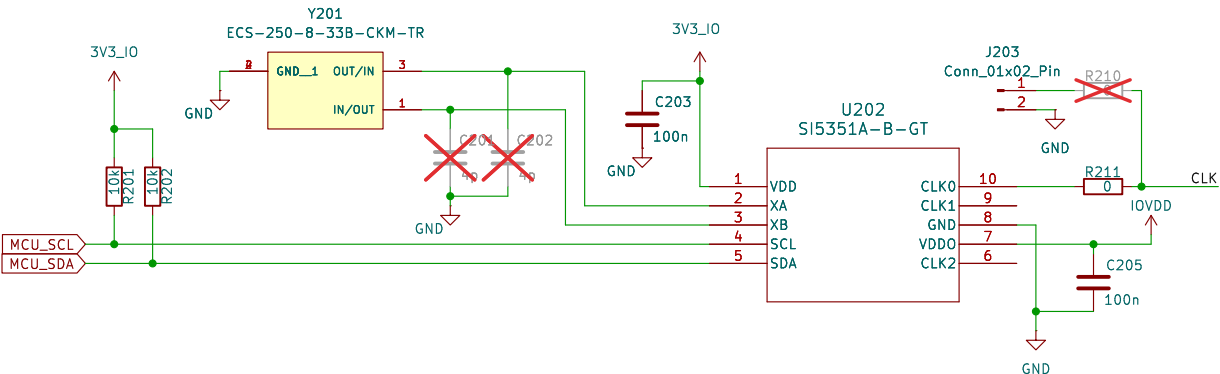


SD card

High resistance needed because of level shifters



Clock generator



Roope Keskinen / roope.r.keskinen@tuni.fi

TAU

Sheet: /SysCtrl/
File: SysCtrl.kicad_sch

Title: Didactic PCB

Size: A3 Date: 2026-03-13
KiCad E.D.A. 9.0.6

Rev:
Id: 2/7

TAU

Title: *Didactic PCB*

Size: A3	Date: 2026-03-13
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KiCad E.D.A. 9.0.6

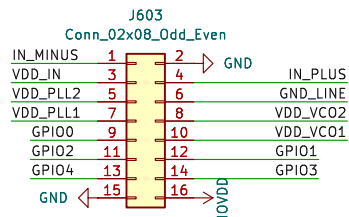
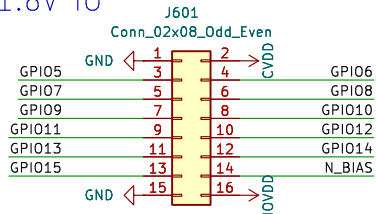
Rev:
Id: 3/7

PMOD IO

U201B

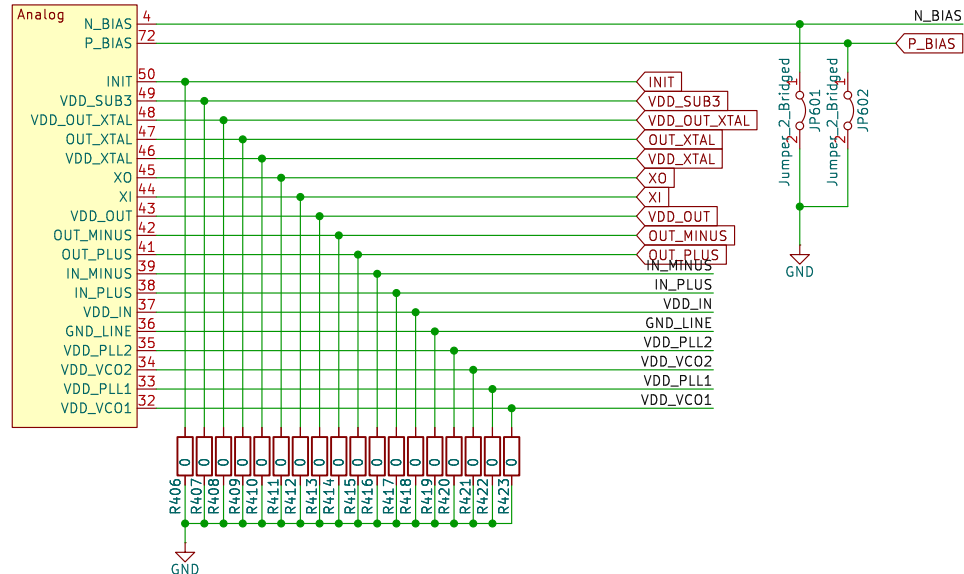
PMOD	GPI00	30	GPI00
	GPI01	28	GPI01
	GPI02	27	GPI02
	GPI03	26	GPI03
	GPI04	25	GPI04
	GPI05	18	GPI05
	GPI06	17	GPI06
	GPI07	16	GPI07
	GPI08	14	GPI08
	GPI09	13	GPI09
	GPI010	12	GPI010
	GPI011	10	GPI011
	GPI012	9	GPI012
	GPI013	7	GPI013
	GPI014	6	GPI014
	GPI015	5	GPI015

1.8V IO

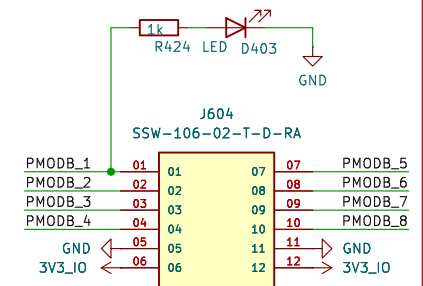
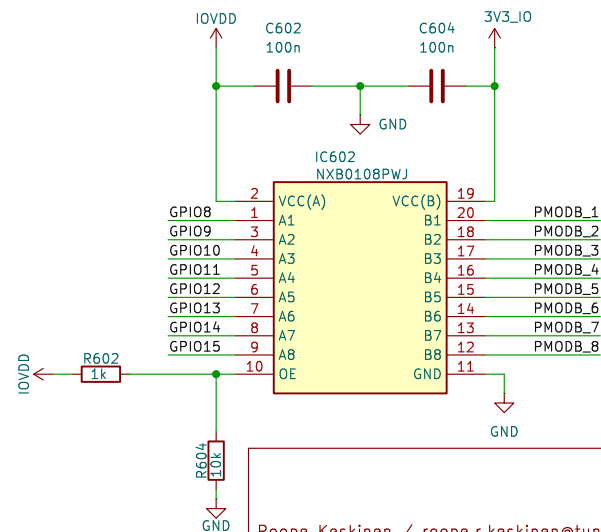
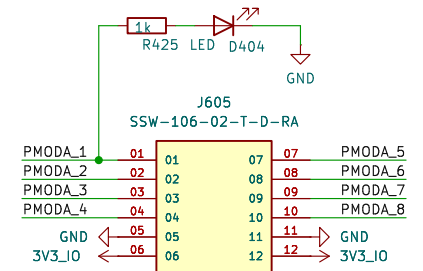
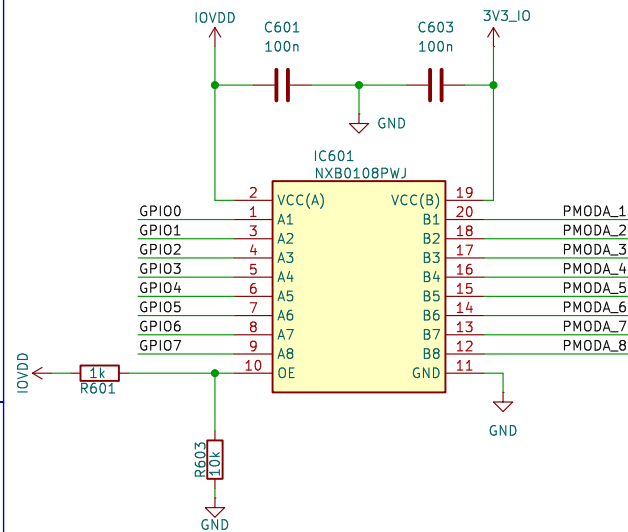


Analog

U201C



1.8V/3.3V Level shifters and PMOD connectors



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TAU

Sheet: /PMOD/

File: PMOD.kicad_sch

Title: Didactic PCB

Size: A4 Date: 2026-03-13

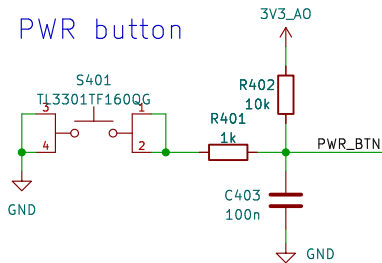
KiCad E.D.A. 9.0.6

Rev:

Id: 4/7

Power management microcontroller

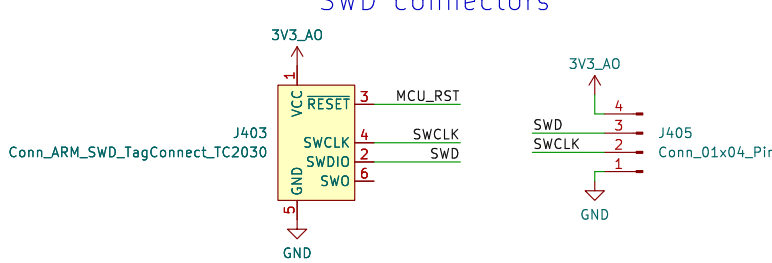
PWR button



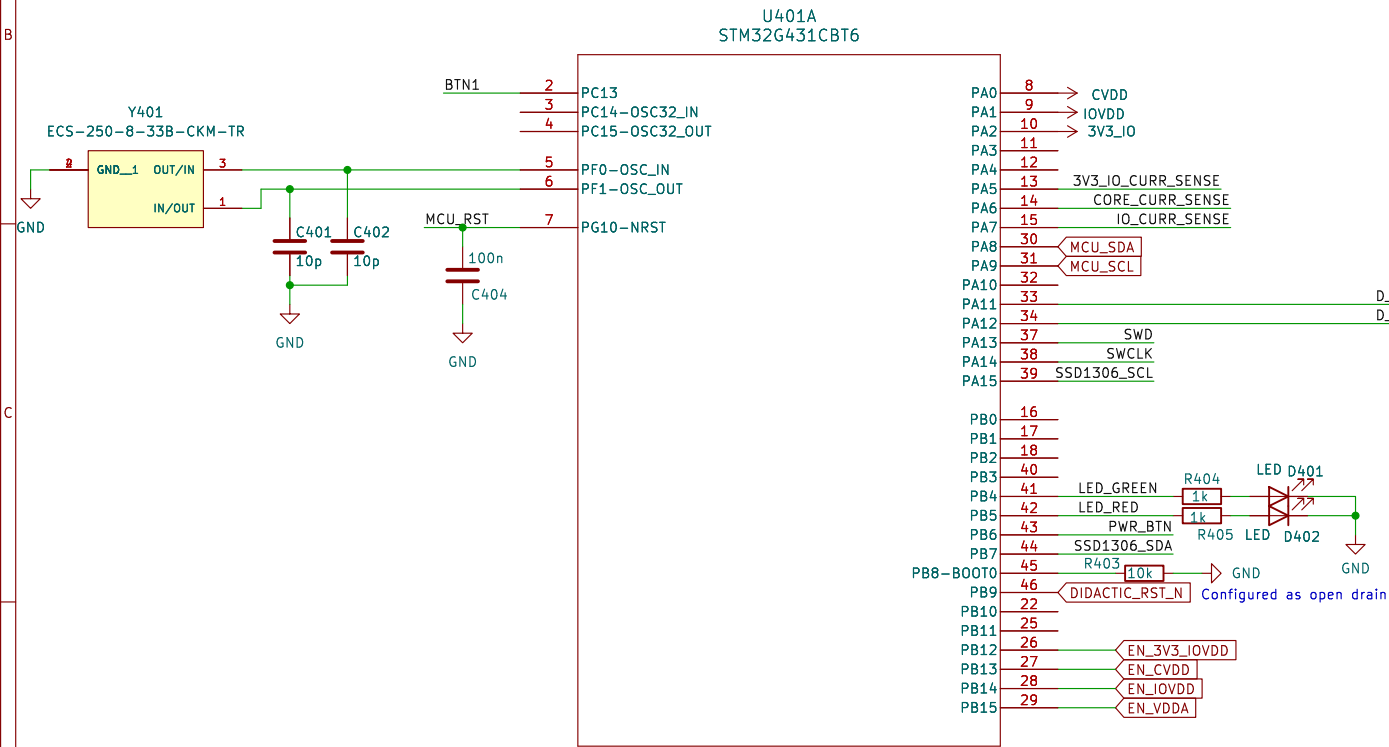
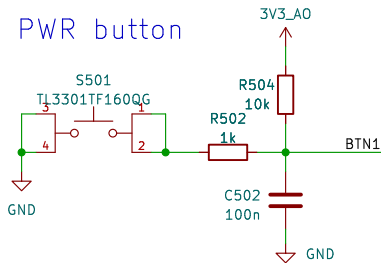
Connector optional SSD1306



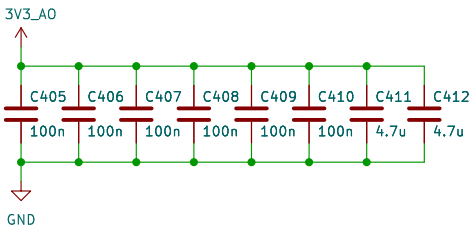
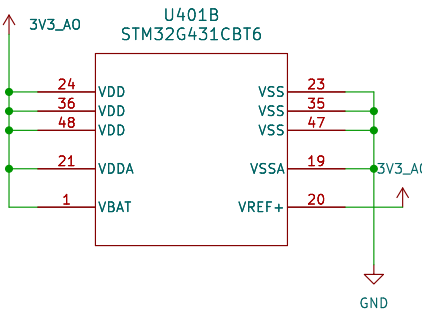
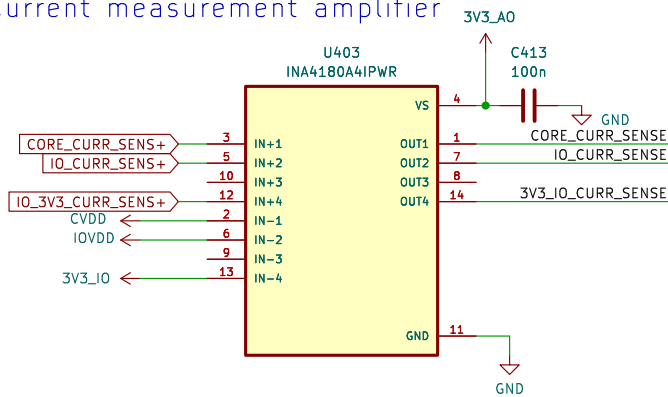
SWD connectors



PWR button



Current measurement amplifier



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TAU

Sheet: /PowerMCU/

File: PowerMCU.kicad_sch

Title: Didactic PCB

Size: A3

Date: 2026-03-13

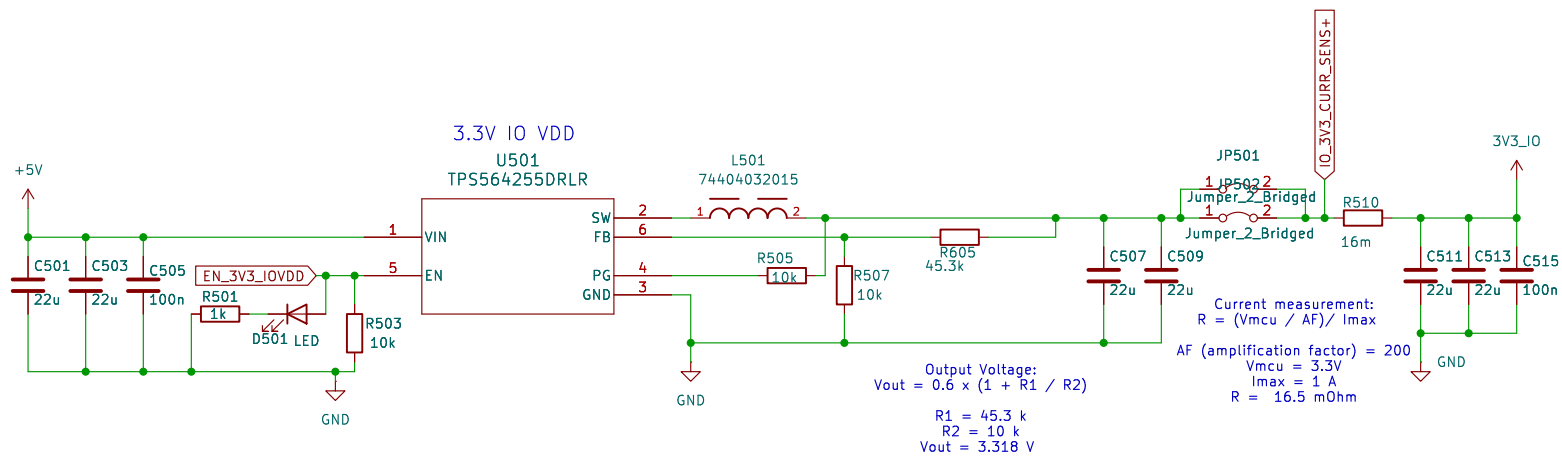
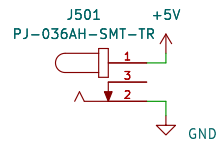
Rev:

KiCad E.D.A. 9.0.6

Id: 5/7

Power SMPS

Power-up sequence:
0. 3.3V (Always on), RST low
PWR button pressed
1. IO 1.8V, IO 3.3V
2. Core 0.9V
3. Enable clock
4. RST high



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TAU

Sheet: /Power_SMPS/

File: power.kicad_sch

Title: Didactic PCB

Size: A4

Date: 2026-03-13

Rev:

KiCad E.D.A. 9.0.6

Id: 6/7

Always on 3V3

TALL

51

File: untitled.kicad.pcb

TH	RI	II	RO
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[illegible]

KIC 15

[illegible]

11. 7.17

8